RAJALAKSHMI ENGINEERING COLLEGE

NABEELA MEERAL M 2024-AIDS N2

Started on	Friday, 24 October 2025, 10:42 AM
State	Finished
Completed on	Tuesday, 4 November 2025, 11:14 AM
Time taken	11 days
Marks	1.00/1.00
Grade	**4.00** out of 4.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    int duplicate = -1;
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < i; j++)
        {
            if (arr[i] == arr[j])
            {
                duplicate = arr[i];
                break;
            }
        }
        if (duplicate != -1) break;
    }
    printf("%d\n", duplicate);
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

NABEELA MEERAL M 2024-AIDS N2

Started on	Friday, 24 October 2025, 10:44 AM
State	Finished
Completed on	Tuesday, 4 November 2025, 11:20 AM
Time taken	11 days
Marks	1.00/1.00
Grade	**4.00** out of 4.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    int arr[n];
    for(int i = 0; i < n; i++)
        scanf("%d", &arr[i]);
    int slow = arr[0];
    int fast = arr[0];
    do
    {
        slow = arr[slow];
        fast = arr[arr[fast]];
    } while (slow != fast);
    slow = arr[0];
    while(slow != fast)
    {
        slow = arr[slow];
        fast = arr[fast];
    }
    printf("%d\n", slow);
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

NABEELA MEERAL M 2024-AIDS

N2

Started on	Friday, 24 October 2025, 10:48 AM
State	Finished
Completed on	Tuesday, 4 November 2025, 11:36 AM
Time taken	11 days
Marks	1.00/1.00
Grade	**30.00** out of 30.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int T;
    scanf("%d", &T);
    while(T--)
    {
        int N1, N2;
        scanf("%d", &N1);
        int arr1[N1];
        for(int i = 0; i < N1; i++)
            scanf("%d", &arr1[i]);
        scanf("%d", &N2);
        int arr2[N2];
        for(int i = 0; i < N2; i++)
            scanf("%d", &arr2[i]);
        for(int i = 0; i < N1; i++)
        {
            for(int j = 0; j < N2; j++)
            {
                if(arr1[i] == arr2[j])
                {
```

```
                {
                    printf("%d ", arr1[i]);
                    break;
                }
            }
        }
        printf("\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

NABEELA MEERAL M 2024-AIDS

N2

Started on	Friday, 24 October 2025, 10:51 AM
State	Finished
Completed on	Tuesday, 4 November 2025, 11:50 AM
Time taken	11 days
Marks	1.00/1.00
Grade	**30.00** out of 30.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int T;
    scanf("%d", &T);
    while(T--)
    {
        int N1, N2;
        scanf("%d", &N1);
        int arr1[N1];
        for(int i = 0; i < N1; i++)
            scanf("%d", &arr1[i]);
        scanf("%d", &N2);
        int arr2[N2];
        for(int i = 0; i < N2; i++)
            scanf("%d", &arr2[i]);
        int i = 0, j = 0;
        while(i < N1 && j < N2)
        {
            if(arr1[i] < arr2[j])
            {
                i++;
```

```
                i++;
            }
            else if(arr1[i] > arr2[j])
            {
                j++;
            }
            else
            {
                printf("%d ", arr1[i]);
                i++;
                j++;
            }
        }
        printf("\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

NABEELA MEERAL M 2024-AIDS

N2

Started on	Friday, 24 October 2025, 10:53 AM
State	Finished
Completed on	Friday, 24 October 2025, 11:05 AM
Time taken	11 mins 50 secs
Marks	1.00/1.00
Grade	**4.00** out of 4.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    int A[n];
    for(int i = 0; i < n; i++)
        scanf("%d", &A[i]);
    int k;
    scanf("%d", &k);
    int found = 0;
    for(int i = 0; i < n; i++)
    {
        for(int j = i + 1; j < n; j++)
        {
            if(A[j] - A[i] == k)
            {
                found = 1;
                break;
            }
        }
        if(found)
            break;
    }
    printf("%d\n", found);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔

	Input	Expected	Got	
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

NABEELA MEERAL M 2024-AIDS N2

Started on	Friday, 24 October 2025, 10:57 AM
State	Finished
Completed on	Tuesday, 4 November 2025, 12:05 PM
Time taken	11 days 1 hour
Marks	1.00/1.00
Grade	**4.00** out of 4.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    int A[n];
    for(int i = 0; i < n; i++)
        scanf("%d", &A[i]);
    int k;
    scanf("%d", &k);
    int i = 0, j = 1;
    int found = 0;
    while(i < n && j < n)
    {
        if(i == j)
        {
            j++;
            continue;
        }
        int diff = A[j] - A[i];
        if(diff == k)
        {
            found = 1;
            break;
        }
        else if(diff < k)
        {
            j++;
        }
        else
        {
            i++;
        }
    }
    printf("%d\n", found);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course